

RESEARCH ON FEATURE EXTRACTION OF ONLINE VIDEO SENTIMENT INFORMATION BASED ON MULTIMODAL FUSION

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Abstract - Network videos contain a wealth of sentiment-related information, and extracting this information can realize the analysis and management of online videos. This study extracted features from three modalities of online videos, namely text, speech, and vision. Moreover, bidirectional gated recurrent unit (BiGRU) was used to extract temporal features. A method combining multihead attention mechanism with maximum pooling (MHA-MP) model was designed. Texts were fused with speech and vision respectively, and then the fused features were spliced with text features to classify online video sentiment information. The Carnegie Mellon University-multi-modal opinion sentiment intensity (CMU-MOSI) and Carnegie Mellon University-multi-modal opinion sentiment and emotion intensity (CMU-MOSEI) datasets were used for verification. Compared with unimodal and bimodal methods, the multimodal fusion approach achieved better performance, obtaining smaller mean absolute errors (MAEs) and larger Pearson correlation coefficients (Corr), accuracy (ACC)-2, F1 values, and ACC-7 on the datasets. Compared to other multimodal fusion methods, the proposed approach exhibited superior performance, with an MAE of 0.695, a Corr of 0.789, and an ACC-7 of 48.1% for the CMU-MOSI dataset, and an MAE of 0.537, a Corr of 0.778, and an ACC-7 of 53.3% for the CMU-MOSEI dataset. These results verify the reliability of the proposed method for analyzing sentiment information in online videos and its potential of practical applications.

Keywords: Text feature, Audio feature, Visual feature, Bidirectional gated recurrent unit, Transformer model.

1. Introduction

With the continuous development of the Internet and the rising spread of various mobile devices, social network platforms such as TikTok, Bilibili, and YouTube have experienced rapid growth and innovation [1]. Network videos have emerged as a significant communication medium for individuals to share their lives and ideas, leading to a surge in online videos. Human sentiment information can be conveyed through expressions and language. Online video contains expressions, words, images, and other information. Analyzing the sentiment information of online videos is an important task, which can aid network managers in understanding the sentiment of subjects, preventing possible extreme emotions, making better video recommendations for users, and enhancing human-machine interaction [2]. Multimodal fusion integrates data from diverse modalities to improve analytical results and has

found wide application in various domains [3], such as medical diagnosis [4] and land monitoring [5]. Jigyasu et al. [6] employed the serial-based feature fusion technique to fuse features from different networks for fault diagnosis of rotating machinery, achieving a remarkable efficiency rate of 99.7% in experiments. Nebili et al. [7] developed a multimodal fusion encoding method for infrared image target recognition and compared the fusion effects of different levels. The experiment found that the method demonstrated classification accuracy of 96.96% and 71.26% on different datasets. Jian et al. [8] constructed a dataset for detecting false restaurant reviews. Integrating metadata features and text features, they achieved a 93.12% accuracy rate for false review detection through a multi-layer perceptron with an attention mechanism. Zandigohar et al. [9] proposed a Bayesian feature fusion framework, integrating electromyogram features and visual features for inferring the

grasping intention of mechanical prosthetic hands. It was found that the accuracy rate of the fused features reached 95.3%, which was significantly better than that of a single feature modality. Shaikh et al. [10] fused the image representation of audio signals and the spatial information in the video modality for multimodal human action recognition.

The experiments found that the system achieved accuracies of 87.9% and 79.0% respectively on the UCF51 and Kinetics Sounds datasets. In the analysis of sentiment information, integrating information of multiple modalities can enable a more comprehensive understanding of a person's emotional state [11]. Therefore, multimodal fusion approaches have also been employed in sentiment information analysis [12]. Al-Tameemi et al. [13] introduced a visual text multimodal fusion (MMF) model and found it have superior performance across four real-world datasets. Wang et al. [14] utilized the multi-attention long short-term memory (LSTM) for multimodal information fusion, achieving a performance boost of 5-10% compared to existing models on benchmark datasets. Zhao et al. [15] proposed a text low-rank fusion-based multimodal emotion recognition network, taking advantage of the complementarity between different modalities. Through a large number of ablation experiments, the effectiveness of this model was proved. Yilmaz et al. [16] proposed a multimodal fusion approach of electroencephalogram (EEG) and electrooculogram (EOG) for emotion recognition. After fusing the EEG-EOG features, a support vector machine was employed for classification. Experiments were conducted on the multimodal DEAP dataset, proving the classification performance of this method. Middya et al. [17] fused audio and video data to establish a multimodal emotion recognition model, achieving high accuracy rates of 99% and 86% respectively on the SAVEE and RAVDESS datasets. From the perspective of existing studies, in multimodal fusion, most of them focus on the fusion of two modalities, and there are relatively few fusions of more than two modalities. Therefore, this study extracted text, speech, and visual features based on multimodal fusion for online video sentiment analysis. A method combining multihead attention mechanism with maximum pooling (MHA-MP) was designed, and its performance was verified through experiments on datasets. This article offers a novel and practical method for extracting, fusing, and categorizing sentiment information in online

videos, providing theoretical references for further research on multimodal fusion.

2. Sentiment Information Analysis based on Multimodal Fusion

2.1 Extraction of Text, Audio, and Visual Features

Online videos contain sentiment information conveyed through text, audio, and visual elements. This paper primarily focuses on feature extraction from these three modalities. Beginning with text, as computers cannot directly interpret text, extracting sentiment information from text requires vectorized encoding. Commonly used methods include word-to-vector (Word2vec), Glove, etc. Bidirectional encoder representation from transformers (BERT), a Transformer-based technique [18], can capture contextual dependencies and resolve word ambiguity, making it highly effective in natural language processing. Therefore, for sentence $T = (\omega_1, \omega_2, \dots, \omega_n)$, this paper chooses BERT to realize extraction of text sentiment information:

$$M_t = BERT(\omega_i), \quad (1)$$

where $M_t \in R^{T_t \times d_t}$, T_t is the text sequence's length, and d_t is the feature size of the text sequence. The speech sentiment information features extracted by BERT are 768 dimensions.

In the feature extraction of speech emotion information, many open-source tools exist, such as OpenSMILE [19] and Librosa [20]. For original speech A , COVAERP [21] is chosen to realize the extraction of speech emotion information:

$$M_a = COVAERP(A), \quad (2)$$

where $M_a \in R^{T_a \times d_a}$ (T_a is the length of the speech sequence, and d_a is the feature size of the speech sequence). The speech emotion information features extracted by COVAERP include 12-dimensional Mel-frequency cepstral coefficient (MFCC), pitch tracking, etc., totaling 74 dimensions.

For feature extraction of visual sentiment information, this paper uses the Facet library [22]:

$$M_v = Facet(V), \quad (3)$$

where $M_v \in R^{T_v \times d_v}$ (T_v is the length of the video sequence, and d_v is the feature size of the video sequence). The speech sentiment information extracted from the Facet library includes facial action units and facial poses of the facial action coding system.

2.2 Time-sequence Feature Extraction based on Bidirectional Gated Recurrent Unit

As different modalities are collected through varying methods, the features extracted in the previous section are aligned using the P2FA tool [23] to ensure that each modality is equal in time sequence. Subsequently, time-sequence features in the sequence are extracted based on bidirectional gated recurrent unit (BiGRU). GRU, a variant of recurrent neural networks (RNNs) [24], addresses the issues of gradient vanishing and exploding encountered in traditional RNNs. Moreover, GRU's straightforward structure enhances computational efficiency. GRU achieves information selection and processing through update gate z_t and reset gate r_t . For the input value of the current moment (x_t) and the hidden state of the previous moment (h_{t-1}), the calculation process of GRU can be written as:

$$r_t = \sigma(W_r x_t + U_r h_{t-1}), \quad (4)$$

$$z_t = \sigma(W_z x_t + U_z h_{t-1}), \quad (5)$$

$$\tilde{h}_t = \tanh(r_t * U_{\tilde{h}} h_{t-1} + W_{\tilde{h}} x_t), \quad (6)$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t \quad (7)$$

where $W_r, W_z, W_{\tilde{h}}, U_r, U_z, U_{\tilde{h}} \in R^{d \times d}$ are weight matrices and $\tilde{h}$ is the candidate hidden state. The traditional GRU can only capture unidirectional information. In order to be able to obtain bi-directional information, this paper uses the BiGRU structure, i.e., a forward GRU and an inverse GRU are obtained as the corresponding hidden layer representations, respectively, and the time-sequence features extracted after BiGRU processing are:

$$X_t = \text{BiGRU}(M_t), \quad (8)$$

$$X_a = \text{BiGRU}(M_a), \quad (9)$$

$$X_v = \text{BiGRU}(M_v), \quad (10)$$

where $X_t \in R^{T_t \times d_k}$, $X_a \in R^{T_a \times d_k}$, and $X_v \in R^{T_v \times d_k}$ (d_k is the unified feature size).

2.3 Multimodal Fusion and Classification

Simple feature concatenation is difficult to capture correlations between modalities. To address this challenge, this paper leverages the Transformer model [25] and introduces a method combining multihead attention mechanism with maximum pooling (MHA-MP). The structure of this method is outlined in Figure 1.

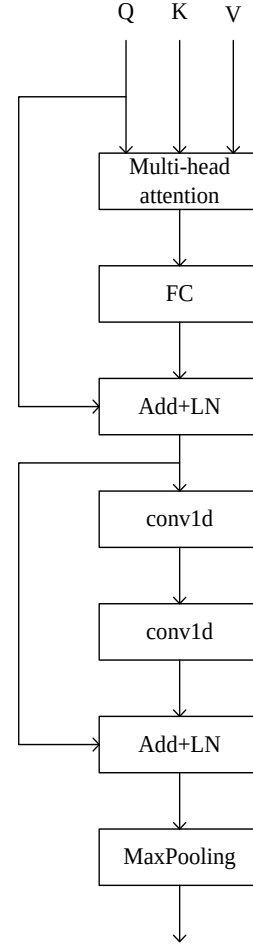


Figure 1: MHA-MP structure (FC: fully connected layer; conv1d: one-dimensional convolutional layer; Add: skip connection; LN: layer normalization)

As shown in Figure 1, in multimodal fusion, in order to simplify the model while obtaining the correlation information between modalities, this paper fuses text with speech and vision separately and then splices the fused features and text features again as the final multimodal fusion features to realize the classification of network video emotion information. Taking the fusion of text and speech as an example, speech modality H^a is regarded as query Q , and text modality H^t is regarded as key value (KV). The related calculations are:

$$Q = \max(0, H^t W_a + b_a), \quad (11)$$

$$K = V = \max(0, H^t W_t + b_t), \quad (12)$$

where W_a and W_t are weights, b_a and b_t are bias. The calculation process of the multi-head attention is:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), \quad (13)$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (14)$$

$$\text{MultiHead}(Q, K, V) = \text{concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h), \quad (15)$$

where W_i^Q , W_i^K , and W_i^V are parameter matrices. After the computation results of h attention heads are obtained, they are inputted to the fully connected layer for feature transformation. After skip connection and layer normalization, the initial fusion feature of text and speech is obtained. Then, the feature mapping is performed using two one-dimensional convolutional layers. The final text and speech fused feature f_{ta} obtained again after skip connection and layer normalization. Redundant information is eliminated through maximum pooling:

$$f'_{ta} = \text{MaxPooling}(f_{ta}). \quad (16)$$

The text and vision fused feature f'_{tv} is obtained using the same method. It is spliced with the feature vector h_n^t of the last time step of text:

$$f_{fuse} = \text{concat}(f'_{ta}, f'_{tv}, h_n^t). \quad (17)$$

The finally obtained multimodal fusion feature f_{fuse} is input into the softmax layer to complete the classification of online video sentiment information.

3. Experiments and Results

3.1 Experimental Dataset

The experimental datasets were CMU-MOSI [26] and CMU-MOSEI [27]. The CMU-MOSI dataset included 93 videos from YouTube, and they were divided into 2,199 short video clips. Seven sentiment categories were included, ranging from strongly negative to strongly positive, with a linear scale of -3 to +3. The CMU-MOSEI dataset included 23,453 video clips from online video sites, and the sentiment tagging was the same as the CMU-MOSEI dataset. The sample distribution of the two datasets is presented in Table 1.

Table 1. Experimental datasets

	Train	Valid	Test
CMU-MOSI	1,284	229	686
CMU-MOSEI	16,326	1,871	4,659

For the CMU-MOSEI dataset, the features extracted from BERT, COVAERP, and Facet were 768-dimensional, 74-dimensional, and 47-dimensional, respectively.

Similarly, for the CMU-MOSI dataset, the features extracted from BERT, COVAERP, and Facet were 768-dimensional, 74-dimensional, and 35-dimensional, respectively.

For the above two datasets, there were five evaluation indicators for multimodal sentiment information analysis: mean absolute error (MAE), Pearson correlation coefficient (Corr), binary-classification accuracy (ACC-2), seven-classification accuracy (ACC-7), and F1 score. There are two calculation methods of ACC-2 and F1 scores:

① The original sentiment labels were divided into negative [-3,0) and non-negative [0,3], and neutral label 0 is included in the non-negative label.

② The original sentiment labels were divided into negative [-3,0) and positive [0,3], and neutral label 0 was excluded.

-/- was used for representation in the results. The left was corresponding to the results calculated in ①, and the right was corresponding to the results calculated in ②.

3.2 Analysis of Results

In order to determine the performance of the multimodal fusion method designed in this paper for analyzing the sentiment information of online video, different modalities were used as inputs for comparison:

- (1) T: use text only,
- (2) A: use voice only,
- (3) V: use vision only,
- (4) T+A: using text + speech,
- (5) T+V: using text + vision,
- (6) V+A: using speech + vision,
- (7) T+A+V: the multimodal fusion approach designed in this paper.

Evaluations were performed using MAE, Corr, and ACC-7 (Table 2).

As shown in Table 2, in the unimodal comparison, the classification of sentiment information in online videos performed the worst when utilizing only visual features, followed by speech features, while the single text modality approach exhibited the best performance, achieving a smallest mean absolute error (MAE) of 0.733, a highest Corr of 0.742, and a maximum ACC-7 of 45.2% on the CMU-MOSI dataset. Moreover, in the comparison of two modalities, T+A/T+V significantly outperformed V+A. This result can be attributed to the fact that the single speech and video modalities are less effective than text when classifying sentiment information. It verified the credibility of fusing text with the other two modalities in multimodal fusion. Lastly, the multimodal fusion approach (T+A+V) yielded the best performance on both datasets. This outcome demonstrates multimodal fusion's effectiveness in analyzing sentiment information in online videos.

Table 2. Comparison of results for different modes

		T	A	V	T+A	T+V	V+A	T+A+V
CMU-MOSI	MAE	0.733	0.807	0.812	0.721	0.719	0.833	0.695
	Corr	0.742	0.633	0.625	0.745	0.751	0.656	0.789
	ACC-7	45.2	39.6	38.9	46.1	46.5	40.1	48.1
CMU-MOSEI	MAE	0.645	0.711	0.732	0.642	0.635	0.721	0.537
	Corr	0.737	0.651	0.6455	0.755	0.761	0.675	0.778
	ACC-7	50.7	42.1	42.3	51.6	51.8	48.9	53.3

Then, to further validate the effectiveness of the proposed method, the method of BiGRU-based time-sequence feature extraction and MHA-MP-based multimodal fusion was compared with other multimodal fusion methods, including:

- (1) the tensor fusion network (TFN) [28],
- (2) the low-rank multimodal fusion (LMF) model [29],
- (3) the multimodal transformation fusion network (MulT) [30],
- (4) the interactive canonical correlation network (ICCN) [31],
- (5) modality-invariant and specific representations for multimodal sentiment analysis (MISA) [32].

The results for the CMU-MOSI dataset are presented in Table 3. Table 3 illustrates that the performance of the TFN and LMF methods in analyzing online video sentiment information was poor.

The MAE values obtained for both methods were higher than 0.9, and the Corr values were lower than 0.7, and the ACC-7 values were 34.9% and 33.2%, respectively, which were low. The performance of the MulT and ICCN methods showed improvement compared to the TFN and LMF methods. The MAE and Corr for MISA were 0.804 and 0.764, respectively, which were superior to the previous four methods; however, its ACC-2 and F1 score were inferior to those of the MulT and ICCN methods. Ultimately, the method proposed in this paper achieved an MAE of 0.695, showing a decrease of 0.109 compared to the MISA method, and a Corr of 0.789, showing an increase of 0.025 compared to the MISA method. The ACC-2 and F1 score were superior, while the ACC-7 was 48.1%, indicating a 9.1% increase compared to the ICCN method. These outcomes demonstrate that the method presented in this paper exhibited the best performance for the CMU-MOSI dataset.

Table 3. Comparison of results for the CMU-MOSI dataset

	TFN	LMF	MulT	ICCN	MISA	The method proposed in this paper
MAE	0.901	0.917	0.861	0.862	0.804	0.695
Corr	0.698	0.695	0.711	0.714	0.764	0.789
ACC-2	-/80.8	-/80.8	81.5/84.1	-/83.0	80.79/82.10	84.4/86.2
F1	-/80.7	-/80.7	80.6/83.9	-/83.0	80.77/82.03	84.6/86.4
ACC-7	34.9	33.2	-	39.0	-	48.1

The results for the CMU-MOSEI dataset are displayed in Table 4.

Table 4. Comparison of results for the CMU-MOSEI dataset

	TFN	LMF	MulT	ICCN	MISA	The method proposed in this paper
MAE	0.593	0.623	0.580	0.565	0.568	0.537
Corr	0.700	0.677	0.703	0.713	0.724	0.778
ACC-2	-/82.5	-/82.0	-/82.5	-/84.2	82.59/84.23	84.3/85.6
F1	-/82.1	-/82.1	-/82.3	-/84.2	82.67/83.97	84.1/85.7
ACC-7	50.2	48.0	-	51.6	-	53.3

Table 4 shows that the MAE values for the TFN and LMF methods were relatively high at 0.593 and 0.623, respectively, and their ACC-7 was 50.2% and 48.0%, respectively. The ACC-7 of the ICCN was 51.6%. The MAE, Corr, ACC-2, and F1 score for the MISA method were 0.568, 0.724, 82.59%/84.23%, and 82.67%/83.97%, and its F1 score was slightly lower than the ICCN method.

The method proposed in this paper achieved a lowest MAE of 0.537, a highest Corr of 0.778, an ACC-2 of 84.3%/85.6%, an F1 score of 84.1%/85.7%, and an ACC-7 of 53.3% for the CMU-MOSEI dataset, and its ACC-7 was 1.7% higher than the ICCN method. These results highlight the efficacy of the method proposed in this paper for analyzing sentiment information in online videos.

4. Conclusions

This paper proposed a method fusing text, speech, and visual features for analyzing sentiment information in online videos. The experimental results for the CMU-MOSI and CMU-MOSEI datasets demonstrated that this fusion approach effectively compensated for the limitations of unimodal approaches, yielding improved classification outcomes. The MAE for the CMU-MOSI dataset was 0.695, the Corr was 0.789, and the ACC-7 was 48.1%. The MAE for the CMU-MOSEI dataset was 0.537, the Corr was 0.778, and the ACC-7 was 53.3%. Furthermore, the method outperformed other multimodal fusion techniques, indicating its reliability. The method can be applied for analyzing sentiment information in real online videos to support video classification and advertisement recommendations. In future research, it will be considered how to further improve the quality of multimodal feature extraction, and more modalities of features, such as body posture, will be considered to be added to further enhance the performance of sentiment information analysis of online videos.

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